

CARE HACK'26

Problem statement ID – AI09

Title - AI-Based Sign Language Translator

Team Name – Dream Coders

Team Lead Name - Rohini R

Domain/Theme- Artificial Intelligence/Assistive Technology

College Name- Saranathan College Of Engineering

TEAM MEMBERS

Member 1 - Rohini R

Member 2 - Rajalakshmi R

Member 3 - Poojitha Shree R

Member 4 - Madhav Padmesh S

PROBLEM STATEMENT

Problem:

Communication between hearing-impaired individuals and people who do not understand sign language is a major barrier.

Explanation:

- Over 70 million deaf people rely on sign language globally
- Most people are not trained in sign language
- Communication barriers affect education, healthcare, and employment
- Existing solutions are expensive or not real-time
- There is a need for an AI-based real-time sign language translation system

AI-Based Sign Language Translator

Proposed Solution:

We propose an AI-powered system that uses computer vision and deep learning to detect sign language gestures through a camera and convert them into text and speech in real time.

Key Features:

- Real-time gesture detection
- AI-based gesture recognition
- Instant text conversion
- Text-to-speech output
- Easy-to-use interface
- Low-cost and scalable solution

TECHNICAL APPROACH

Technologies Used:

- Python
- OpenCV (Image Processing)
- Media Pipe (Hand Detection)
- TensorFlow / PyTorch (AI Model)
- NumPy (Data Processing)
- Text-to-Speech Engine
- Web / Mobile Interface

Methodology & Process:

Capture gesture using camera

Detect hand using Media Pipe

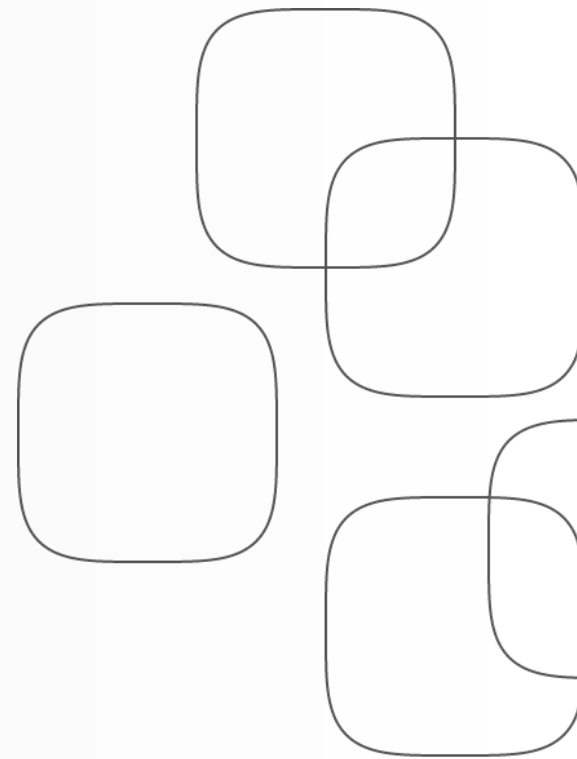
Extract hand landmarks

Process using trained AI model

Classify gesture

Convert into text

Convert text into speech



FEASIBILITY

Feasibility Analysis:

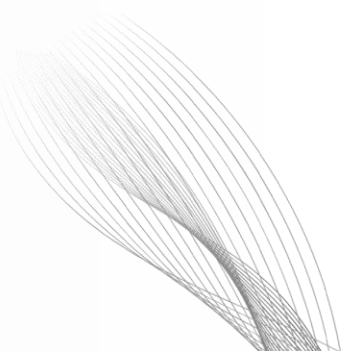
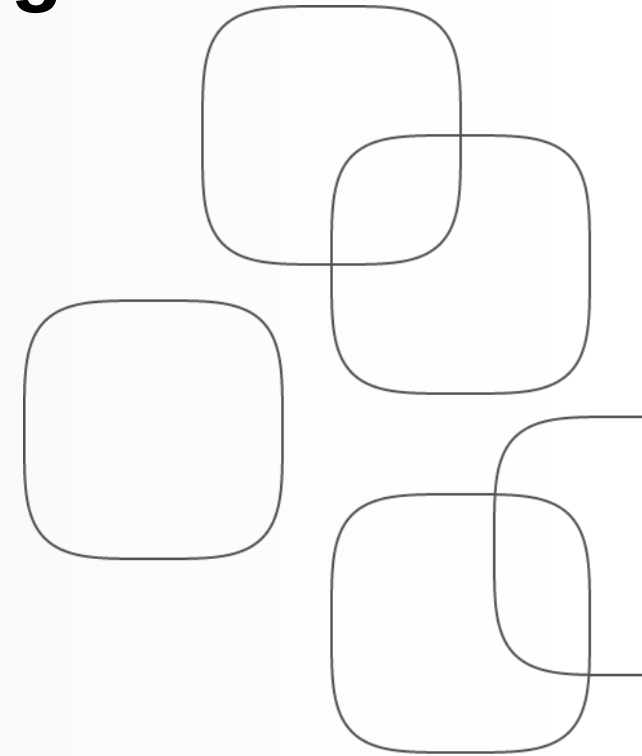
- Uses low-cost hardware like webcam or mobile camera
- Open-source tools reduce development cost
- Can run on laptop or mobile device

Potential Challenges:

- Accuracy of gesture recognition
- Lighting and environmental conditions
- Detection of complex gestures

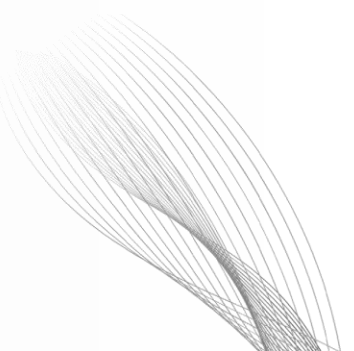
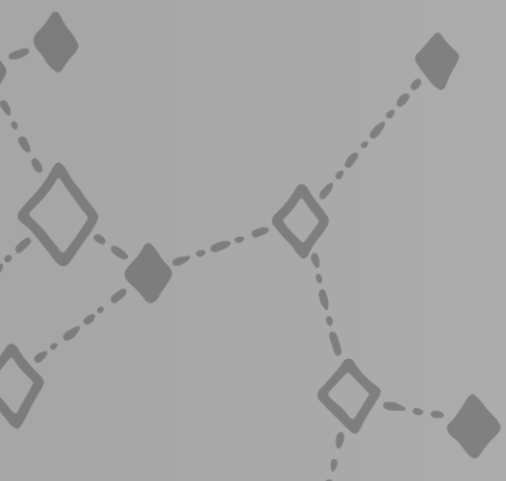
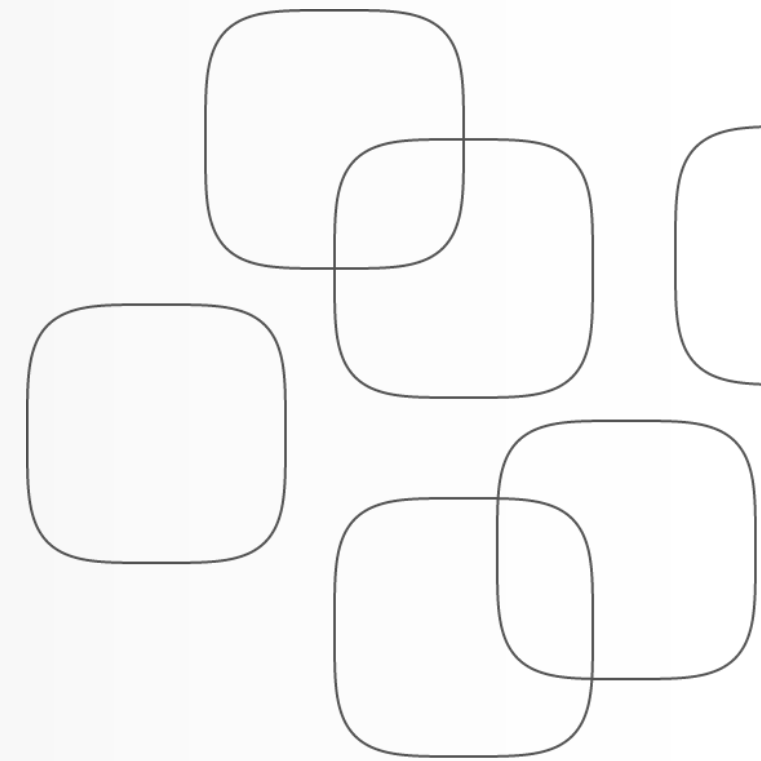
Strategies to Overcome Challenges:

- Train model with large dataset
- Use deep learning optimization
- Improve image preprocessing
- Use better camera quality



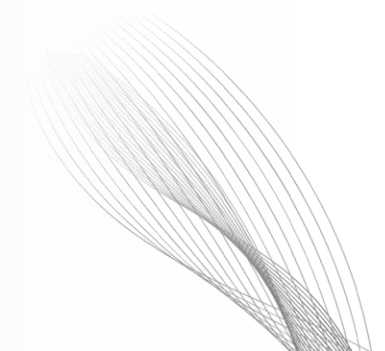
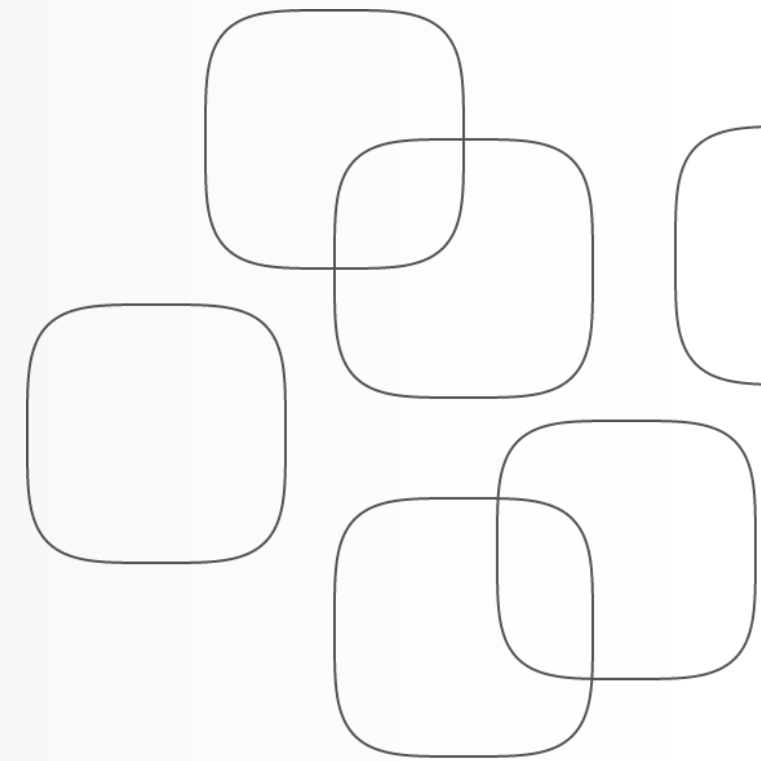
IMPACT ON SOCIETY

- Enables communication for deaf and mute individuals
- Promotes inclusivity and accessibility
- Helps in education and workplaces
- Useful in hospitals and public services
- Improves independence and quality of life
- Supports digital accessibility initiatives



FUTURE ENHANCEMENT

- Full sentence translation
- Speech-to-sign translation
- Mobile application deployment
- Multi-language support
- Integration with wearable devices
- Cloud-based AI system



RESEARCH AND REFERENCES

- Media Pipe Official Documentation
- TensorFlow Official Documentation
- OpenCV Official Website
- IEEE Papers on Sign Language Recognition
- Google AI Research

